Key Theories of Creativity and Innovation

Chapter 1 Day 1 (Wednesday)

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Learning outcomes

By the end of today you will be able to:

- Define creativity and innovation and explain how they differ
- Describe seven theories of creativity and innovation
- Give a workplace example for each theory
- Identify habits and conditions that help or block new ideas
- Compare the theories and see how they complement each other
- Prepare a personal action plan

Plan for the day

Morning session (2 hours)

- Introduction
- Shrinking Earth Theory
- Systems Theory
- Honing Theory
- Everyday Imaginative Thought Theory

Afternoon session (2 hours)

- Evolution Theory
- Investment Theory
- Social Personality Theory
- Bringing the theories together
- My action plan

Warm-up question

Why do some people and organizations keep producing fresh ideas while others repeat the same routines for years?

Think for one minute and share with the person next to you.



What is creativity?

Creativity

The ability to produce ideas, solutions, or products that are both **new** and **useful**.

- A new idea that nobody can use is only a curiosity
- A useful idea copied exactly from elsewhere is only good imitation
- Creativity needs both qualities together

What is innovation?

Innovation

The process of turning a creative idea into something that is put to work and brings value.

- A new service, a better procedure, a new tool, or a new way to organize people
- It must be tested, accepted, and used

In short

Creativity supplies the seed. Innovation grows it into a plant that bears fruit.

Creativity and innovation compared

	Creativity	Innovation
Main question	What new ideas can we imagine?	How do we make the idea work in real life?
Focus	Generating and shaping ideas	Testing, adopting, and spreading ideas
Typical result	A concept, a draft, a plan	A changed service, product, or practice
Typical risk	Ideas that stay on paper	Resistance, cost, and poor planning

Why study theories?

- A theory is an organized explanation
- It helps us notice what we would otherwise miss
- If creativity is a rare gift, we will not encourage it
- If creativity grows from conditions and habits, we will act very differently

Today

Seven theories give us seven lenses. No single lens shows everything, but together they give a rich picture.

Seven theories, seven questions

Theory	The question it answers
1.1.1 Shrinking Earth	How does a connected world push us to create?
1.1.2 Systems	How do the person, the field, and the domain work together?
1.1.3 Honing	How do ideas and thinkers change through refinement?
1.1.4 Everyday Imaginative Thought	Is creativity only for geniuses?
1.1.5 Evolution	How do many ideas compete and the best survive?
1.1.6 Investment	Why do creative people choose unpopular ideas?
1.1.7 Social Personality	How do traits and the social setting shape creativity?

Morning Session

Theories 1.1.1 to 1.1.4

1.1.1 Shrinking Earth Theory

Starting point

Roads, airlines, mobile networks, and the internet make distance matter less. The world feels like a small village.

- Ideas, people, goods, and information travel faster and meet more often
- This meeting of ideas is a source of creativity and a push toward innovation



How a shrinking world encourages creativity

- **Exposure to diversity:** other cultures and habits challenge our usual way
- **Recombination:** new ideas often combine old ideas from different places
- **Competition and rising expectations:** clients know what is possible elsewhere
- **Faster spread:** a good practice can be known by thousands within days
- **Shared problems:** health, food, climate, and education need shared solutions

Example: mobile money

- The old idea of a bank account joined the mobile phone
- Services such as telebirr changed how many Ethiopians pay and send money
- Recombination plus a connected world produced a new practice



Limits and cautions

- Copying a practice without asking if it fits our context is imitation, not innovation
- Too much information can cause confusion
- The digital gap can leave some people behind

The creative person is an adapter

What do we keep? What do we change? What do we add so that the idea works here?

Key ideas: Shrinking Earth

Remember

- A connected world spreads and mixes ideas
- Creativity means adapting borrowed ideas to local needs
- Access and context matter

Activity 1: Idea Import (10 minutes)

Think of a good practice from another office, university, or country

1. What is the practice and where did you see it?
2. What problem in your work could it help solve?
3. What must change so that it fits our situation and culture?

Share with the person next to you.

Check your understanding

Questions

1. Why can a smaller world lead to more creativity?
2. Why is copying an idea from another country not the same as being creative?

A young nurse designs a better patient record sheet

Is she creative?

- Theory developed mainly by Mihaly Csikszentmihalyi
- Creativity is not found only inside a person's head
- It happens in the interaction of three parts of a system



Three parts of the system

Part	Meaning	Example in an office
Domain	Knowledge, rules, and symbols of a field of work	Records standards, regulations, accounting rules
Field	People who decide what is good and accepted	Clinic head, committee, supervisors
Individual	The person who learns the domain and brings a new variation	The nurse, the secretary, you

When is an act creative?

According to this theory

An individual changes the domain **and the field accepts the change**, so it becomes part of the domain.

- A good idea that is never noticed or accepted has not yet become a creative contribution
- Good ideas need good communication and good relationships, not only good thinking

Systems thinking in organizations

- An organization is a connected whole: inputs, activities, outputs, feedback
- A new form in the registrar's office changes the work of the finance office
- A faster approval may put more pressure on the archive
- Think about ripple effects before you act



Key ideas: Systems

Remember

- Domain, field, and individual interact
- Learn the domain and understand how the field judges ideas
- Present your idea in a language the field respects
- Every change has side effects elsewhere

Activity 2: Map the System (15 minutes)

In groups of four

Choose one routine process, such as processing a leave request or receiving a visitor.

1. Draw the steps from first input to final output
2. Mark who must approve or accept at each step (the field)
3. Circle one step to improve and show which other steps it affects

Check your understanding

Questions

1. Name the three parts of the system and give one workplace example of each
2. Why might a good idea fail to be recognized as creative?

Break

10 minutes

A common belief

Creativity is a flash of lightning. Nothing, and then a brilliant idea.

- Studies of writers, scientists, and designers show ideas are shaped slowly
- Honing Theory was developed by the cognitive scientist Liane Gabora



What does “honing” mean?

- To hone is to sharpen gradually
- Creativity is gradual refinement of a **mental model**, our understanding of the problem
- Each draft or trial gives the idea an outer form
- That outer form teaches the creator something new

Two-way change

The idea changes the thinker, and the thinker changes the idea.

Example: improving a visitor register

Round	What the worker does	What is learned
Draft 1	Simple table with name, date, time	People forget the purpose of the visit
Draft 2	Adds purpose and office to be visited	Visitors do not know which office is responsible
Draft 3	Adds a guide and signs at the gate	Guards need a quick way to call the right office
Draft 4	Adds a phone list and instructions for guards	The process is smooth and the worker understands visitors far better

Habits of a good honer

- Start with a rough first version
- Ask for feedback early from the people who will use the result
- Keep earlier versions to see how your thinking grew
- After each version ask: what did this attempt teach me?
- Give the process enough time

Key ideas: Honing

Remember

- Creativity is often gradual refinement, not a sudden flash
- Each version reveals something and changes the creator's understanding
- Feedback, patience, and repeated adjustment are central

Activity 3: Three Drafts (15 minutes)

Choose a small work item

A form, a notice, or a meeting agenda template.

1. Improve it in three rounds of three minutes
2. After each round, swap with your neighbor, who writes one comment
3. Discuss: how did your thinking change between the first and third version?

Check your understanding

Questions

1. What does it mean to hone an idea?
2. How does Honing Theory challenge the belief that creativity is a flash of genius?

Is creativity only for famous inventors and artists?

This theory answers no. Creativity grows out of ordinary human imagination.

- Daydreaming and planning a journey
- Thinking "what if"
- Telling a story
- Cooking with the ingredients we have
- Repairing a tool with whatever is nearby



The Four C model

Level	Meaning	Example
Mini-c	A new personal insight during learning	A trainee suddenly sees how two procedures connect
Little-c	Everyday creativity at work and home	A secretary invents a colour code that speeds up filing
Pro-c	Professional creativity built on expertise	An engineer designs a cheaper water filter for a rural school
Big-C	Rare, historic achievements	Major scientific discoveries and great works of art

Model proposed by James Kaufman and Ronald Beghetto.

Where does most workplace innovation come from?

- Everyone starts with mini-c and little-c
- These are the soil from which higher levels grow
- Most valuable innovation comes from many small improvements by ordinary people
- If you think you are not creative, look again at the clever small things you already do



Tools of everyday imagination

- **"What if" questions:** what if we did this step first? What if the client filled in the form?
- **Reversal:** instead of reducing complaints, ask how to make clients happier
- **Making do with constraints:** limited money, time, or materials force clever solutions
- **Role switching:** see the process as the client, the new employee, or the manager

Blocks are mostly inside us

"I am not creative," fear of being laughed at, and the comfort of routine.

Key ideas: Everyday Imaginative Thought

Remember

- Creativity is part of ordinary thinking
- The Four C model is a ladder everyone can climb
- Small daily improvements add up to real innovation

Activity 4: Five What-Ifs (10 minutes)

Pick one routine task you do every week

Write five "what if" questions about it. Choose your most interesting one and share it.

Example: What if clients could track the status of their request without visiting our office?

Check your understanding

Questions

1. Give one example of mini-c and one of little-c from your own life
2. Why do small everyday improvements matter for an organization?

Morning recap

1. The world is connected, so ideas travel and combine
2. Creativity needs the domain, the field, and the individual
3. Ideas are honed through repeated drafts
4. Everyone has everyday imagination

Before lunch

Write down one idea from this morning that you want to try this week.

Afternoon Session

Theories 1.1.5 to 1.1.7 and Synthesis

Welcome back: quick review (10 minutes)

Without looking at your notes

Write one sentence on each of the four theories from the morning.

- Shrinking Earth
- Systems
- Honing
- Everyday Imaginative Thought

Compare with a partner and correct each other.

1.1.5 Evolution Theory

- In nature, living things vary
- Helpful variations survive and are passed on, others disappear
- Donald Campbell (1960) proposed that creative thinking follows a similar pattern
- Dean Keith Simonton later developed a Darwinian view of creativity

Blind variation and selective retention



Three stages

Stage	Key question	At work
Variation	How many different ideas can we produce?	Brainstorming, several designs, suggestions from all staff
Selection	Which ideas fit our needs best?	Clear criteria, small pilots, user feedback
Retention	How do we keep and spread what works?	Written procedure, staff training, standard practice

Why “blind”?

- It does not mean careless thinking
- It means nobody can know in advance which idea will succeed
- So **many ideas increase the chance of a good one**
- Highly creative people produce a lot, with many successes and many failures

A fair criticism

Creators are guided by knowledge and judgment. So use the theory as a reminder to generate and select, and not to judge too early.

Application in organizations

1. Staff submit suggestions
2. A team selects the most promising
3. A small pilot tests them
4. A successful pilot becomes policy

Failed pilots are not punished. They are the natural cost of learning.



Key ideas: Evolution

Remember

- Produce varied ideas, select the fittest, retain what works
- No one can predict which idea will succeed
- Separate idea generation from idea evaluation

Activity 5: Idea Tournament (15 minutes)

In groups

1. 5 minutes: list as many ideas as possible to improve service to students or visitors
2. Select the best three using three criteria: **useful, feasible, affordable**
3. Describe in two sentences how to test the winner on a small scale

Check your understanding

Questions

1. What do variation, selection, and retention mean in creative work?
2. Why is it unwise to criticize ideas too early in a brainstorming session?

Robert Sternberg and Todd Lubart

Creative people **buy low and sell high** in the world of ideas.

- They choose ideas that are unknown, unpopular, or doubted
- They develop the idea and persuade others
- When the idea is valued, they "sell high" and move to the next



Defying the crowd

- A new idea looks strange at first
- The crowd prefers what is familiar
- Creative people often meet resistance

Creativity is partly an act of defying the crowd while staying realistic

Six resources (part 1)

Resource	Meaning	At work
Intellectual abilities	See problems in new ways, judge which ideas deserve effort, persuade others	Spot a hidden problem and explain it clearly
Knowledge	Enough to move the field forward but not so much that it traps thinking	Know the rules well and ask if a rule still serves its purpose
Thinking styles	Preference for thinking in new ways and setting one's own direction	Enjoy designing better methods

Six resources (part 2)

Resource	Meaning	At work
Personality	Sensible risk taking, persistence, tolerance of uncertainty	Continue even when a first attempt fails
Motivation	Driven by interest in the task itself	Work on improvement because you care about the result
Environment	A setting that supports and rewards new ideas	Leaders listen, give time, and protect those who try

Selling high takes skill

- A new idea needs evidence, clear communication, and patience
- Example: a digital attendance system
  - Collect data from a small trial
  - Show the time saved
  - Explain how colleagues benefit
- Persuasion is part of the investment



Key ideas: Investment

Remember

- Creative people pursue undervalued ideas and help others see their value
- Six resources interact
- Persuasion and persistence matter as much as the idea

Activity 6: Resource Check (15 minutes)

Rate yourself from 1 (very low) to 5 (very high)

Intellectual abilities, knowledge, thinking styles, personality, motivation, environment.

1. Choose the resource you most want to invest in over the next three months
2. Write one concrete step
3. Share with a partner

Check your understanding

Questions

1. What does it mean to buy low and sell high with ideas?
2. Which resource is strongest in you and which needs the most attention?

Break

10 minutes

Are creative people born that way, or does the setting make them creative?

Both matter. Creativity comes from the meeting of personal traits and a supportive social environment.



The personality side

- Research reviews, such as those by Gregory Feist, point to common traits
- Curious and open to new experiences
- Independent in thinking
- Tolerant of uncertainty
- Self-confident
- Persistent when work is difficult

Good news

Many of these traits can be developed through practice and experience of success.

The social side: Teresa Amabile

Three components within a person

Skills in the domain, skills in creative thinking, and **intrinsic task motivation**

- Intrinsic motivation means doing something because it is interesting, enjoyable, or satisfying in itself
- People are usually more creative when motivated by interest and challenge
- Threats, strict surveillance, and heavy competition tend to lower creativity

Supports and blocks

What supports creativity	What blocks creativity
Freedom to decide how to do the work	Constant monitoring and rigid control
Interesting and challenging tasks	Harsh criticism and ridicule
Enough time and basic resources	Unrealistic deadlines without explanation
Encouragement and recognition	Fear of punishment for mistakes
Diverse teams with open talk	Unhealthy competition and distrust

Culture matters

- Where respect for seniority is strong, junior staff may hesitate to share ideas
- Wise leaders create safe ways to speak
  - Short idea meetings
  - Anonymous suggestion channels
  - Thanking people who raise problems



Key ideas: Social Personality

Remember

- Traits and the social environment interact
- Intrinsic motivation is a powerful driver of creativity
- Leaders can build or destroy creativity through the climate they create

Activity 7: Climate Check (15 minutes)

Step by step

1. Alone: list three things in your office that help creativity and three that block it
2. In groups: combine your lists
3. Choose one blocker that you personally can influence
4. Write one action you will take

Check your understanding

Questions

1. What is intrinsic motivation and why does it matter for creativity?
2. Give two ways a leader can improve the climate for creativity

The seven theories are not rivals

Environment

- Shrinking Earth
- Systems
- Social Personality

Process

- Honing
- Evolution

Person

- Everyday Imaginative Thought
- Investment

A creative organization pays attention to all three at once.

What you can do

Theory	Core idea	What you can do
Shrinking Earth	A connected world spreads and mixes ideas	Learn from others and adapt ideas
Systems	Domain, field, and individual interact	Master your field and build support
Honing	Ideas are refined through drafts	Start rough, seek feedback, improve
Everyday Imaginative Thought	Creativity is ordinary thinking	Ask "what if" about daily tasks
Evolution	Variation, selection, retention	Generate many ideas, pilot the best
Investment	Buy low, sell high, six resources	Develop resources and persuade
Social Personality	Traits and climate together	Protect motivation, build a safe climate

Activity 8: My Creativity Commitments (15 minutes)

Complete these sentences. Be specific and realistic.

1. From the theory of _____, I will start doing _____ by _____
2. From the theory of _____, I will stop or change _____ by _____
3. To help my team, I will _____ at our next meeting

Chapter summary

- Creativity produces ideas that are new and useful, innovation puts them into practice
- **Shrinking Earth**: a connected world spreads ideas and pushes us to adapt
- **Systems**: domain, field, and individual work together
- **Honing**: gradual refinement of ideas and of the creator's understanding
- **Everyday Imaginative Thought**: creativity belongs to ordinary people
- **Evolution**: variation, selection, and retention of ideas
- **Investment**: buy low and sell high with ideas, using six resources
- **Social Personality**: traits and the social environment shape creativity

Review questions for homework

1. Explain the difference between creativity and innovation with an example from your office
2. Which theory is most useful for your work, and why?
3. Choose two theories and show how they complement each other
4. Describe an idea at work that was not accepted. Use Systems or Investment Theory to explain why
5. What three things can a leader do tomorrow to strengthen creative thinking?
6. Describe a time you honed an idea through several versions

Further reading

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Thank you

Tomorrow (Thursday): Chapter 2

Questions and reflections